

# Tightening the Operator/Mechanic split — recommendations

Higher accuracy + constant speed + RPS-safe. Ranked by value vs risk. Nothing here touches the live index or breaks legacy.

## Tier 1 — make it tighter AND faster

### ⚡ Cache the side map

- Classify once; cache keyed on doc-count + MAX(rowid).
- Invalidates only when documents change.
- /api/by\_side + counts become O(1) after first build.

Speed: faster, never slower.

### 📄 Confidence + override sidecar

- tm\_side returns confidence: high(code)/med(title)/low(default).
- sides\_override.json lets you PIN any doc to a side.
- One-click fix for the 'uncertain' tail (append-only).

Accuracy: kills the guesswork.

### 📄 Cover/MAC corroboration

- ONLY for docs with no coverage code.
- Peek OCR'd page 1: 'OPERATOR'S MANUAL' -> operator;
- 'MAINTENANCE ALLOCATION CHART' -> mechanic.

Accuracy on the few, no global cost.

## Tier 2 — polish that complements

### 📄 Combined badge + cross-link

- Mark -12/-13/-14 as 'Combined — also on the other side'.
- So nobody thinks a combined manual is misfiled.

Clarity, tiny effort.

### 📄 Operator-side declutter

- De-emphasize parts-request / NSN-dossier CTAs on the
- operator side (still reachable). Presentation-only.

Simpler operator view.

### 📄 Side in deep links

- ?side= on shared links so a link opens the right side.
- Server already accepts side= on search.

Consistency across users.

## Tier 3 — the one you may not have considered

CHAPTER-LEVEL routing inside combined manuals: a -12 has operator chapters AND maintenance chapters on different pages. Use the OCR'd table-of-contents / chapter headers to send an operator to the operator chapters of a -12 (not the whole book). Biggest accuracy leap for combined manuals; reuses OCR you already have; more effort.

Guardrails kept throughout: classification stays pure-regex (no network), cache invalidates on a cheap signal, everything is sidecar/in-memory (never writes the 3.65 GB index, R1/R6), and all UI stays ES5 for the RPS/legacy build.